

Type of AI	Key features	Use cases
Explainable AI	- Provides transparent explanations for AI decisions - Helps users understand how AI models work - Enhances trust and accountability	Healthcare diagnostics, Fintech - financial risk assessment, legal decisions, regulatory compliance
AI fairness	- Mitigates biases in AI algorithms - Ensures equitable treatment of different groups - Prevents discrimination in AI-driven decisions	Hiring processes, lending and credit decisions, criminal justice system
Emotion AI	- Detects and interprets human emotions - Analyses facial expressions, voice tone, and text sentiment - Enhances human-computer interaction	Customer service, market research, mental health monitoring
Responsive AI	- Adapts and responds to user inputs and context - Improves user experience through natural interactions - Personalised recommendations and services	Chatbots, virtual assistants, recommendation systems, e-commerce
Generative AI	- Creates new content, text, images, or music - Can produce creative and original outputs - Used in content generation and creative tasks	Content generation, art creation, language translation, chatbots

Table 1: What different types of AI do

Fairness and bias: AI models might inherit biases from their training data, resulting in discriminating conclusions. Explainability assists in identifying and correcting these biases, assuring fairness and regulatory compliance.

Regulations and compliance: Many businesses are governed by stringent regulations that demand transparency and responsibility. Explainable AI assists organisations in adhering to legal and ethical requirements.

Human-AI collaboration: To effectively collaborate with AI, humans must grasp AI's reasoning. This is especially true in medical and legal applications.

Techniques for enhancing model interpretability

Model interpretability in the context of explainable AI (XAI) refers to the extent to which humans can comprehend and explain an AI model's predictions and decisions. It involves making complex and frequently inscrutable machine learning models more transparent and understandable, so that users, stakeholders, and experts can gain insight into how the model arrives at

particular outputs or decisions.

Model interpretability seeks to increase transparency in artificial intelligence models. This requires revealing how the model processes input data and produces output predictions. Transparent models are simpler to comprehend and trust. Here are some methods that improve model interpretability.

SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations): These methods assign relevance scores to model elements, making it easier to comprehend their impact on predictions.

Conclusion trees and rule-based models: These present a clear and interpretable collection of the factors that led to a specific conclusion.

Grad-CAM (Gradient-weighted Class Activation Mapping): These techniques highlight the most relevant regions in images that influenced the model's choice.

Local explanations vs global explanations: Depending on the context and requirements, AI models can provide explanations for individual

predictions (local) or for the entire model (global).

Balancing transparency and performance in AI models

Balancing transparency and performance in AI models is crucial for responsible AI development and deployment. Transparency refers to the ability to understand and explain the decision-making process of an AI model, while performance refers to the accuracy and effectiveness of the model in achieving its intended purpose.

Trust and accountability: When users can understand how AI models make decisions, they are more likely to trust and accept those decisions. This is particularly important in high-stakes applications, such as healthcare or criminal justice.

Fairness and bias detection: Transparency allows for the detection and mitigation of biases in AI models. By understanding how models arrive at their conclusions, it becomes possible to identify and address potential sources of bias, ensuring that AI systems operate fairly and impartially.